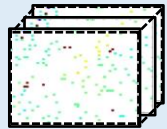


VLM-Driven RF Sensing Framework

Input



$\mathbf{H}_I$

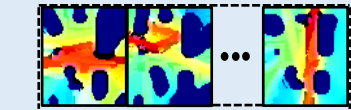
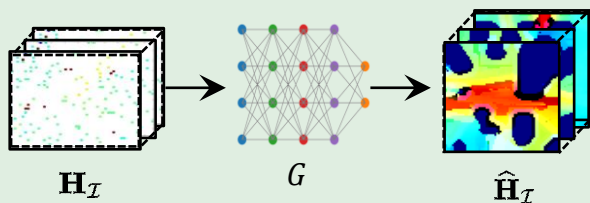
Channel Gains
Measured at
Different Locations



$\mathbf{x}$

Textual Prompt:
Sensing
Requirement

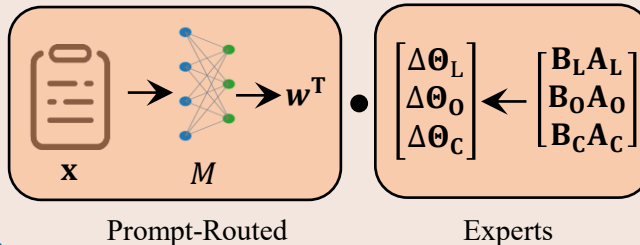
Reconstruction Module ❄️



Estimated Channel Gains in Visual Form

$$\mathbf{H}_I^{\text{Input}} = \left[\left[\hat{\mathbf{H}}_I \right]_{:,1}, \dots, \left[\hat{\mathbf{H}}_I \right]_{:,N_D} \right]$$

Prompt-Routed LoRA Experts (for $\Delta\Theta$) 🔥



Prompt-Routed

Experts

Pre-trained
VLM
(with Θ_P) ❄️



Fine-Tuned VLM

$$\Theta = \Theta_P + \frac{\alpha}{r} \Delta\Theta$$

Probability

Matrix

$\hat{P}_I$

Textual
Caption

$\hat{\mathbf{y}}$

Output



Textual Caption $\hat{\mathbf{y}}$
of Multi-Level
Environmental
Semantics $\hat{\mathbf{I}}$

🔥 Fine-tuning

❄️ Frozen